Name: Jaanhavi Ashwin Kher
SRN: PES2UG24CS100
Date: 8/9/26
SE Banana Problem: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem Statement: Blood Bank Inventory & Emergency Donor Matcher

Architecture Selection

We chose Microservices Architecture for the Blood Bank Inventory & Emergency Donor Matcher System.The system is divided into independent components: Emergency Request Service, Donor Matching Service, Blood Inventory Service, Donor Management Service, Notification Service, and Blood Inventory Database.

Architectural Choice

Microservices Architecture is the most suitable choice because the system requires real-time processing of emergency blood requests, inventory management, donor matching, and rapid notification of eligible donors. Independent services allow different parts of the system to operate and scale separately.

Reason 1: Real-Time Emergency Processing

The Donor Matching Service can independently process emergency requests by checking blood inventory and donor eligibility. The Notification Service can then rapidly send SMS alerts to compatible donors within the required 10 km radius, supporting the requirement for timely emergency notifications.

Reason 2: Scalability and Fault Isolation

The services can be independently deployed and scaled according to demand. During a critical blood shortage, services such as donor matching and notifications can handle increased workloads without requiring the entire system to be scaled. Failure in one service can also be isolated from other services.

Security Advantage

Microservices allow access to sensitive donor and inventory information to be controlled through defined service interfaces and APIs. Only authorized services can access the information they require, helping protect donor information and prevent unauthorized access to blood inventory data.

Performance Benefit

Independent scaling of high-demand services, particularly the Donor Matching Service and Notification Service, improves the system's ability to process emergency requests quickly and send notifications within the required response time. The Blood Inventory Service can also maintain transactional consistency when allocating blood units, preventing simultaneous allocation of the same blood bag.